

Course Code: 23ADL501		Course Title: Intelligent Systems - II Laboratory	
Course Category: Major		Course Level: Higher	
L:T:P(Hours/Week) 0: 0: 4	Credits:2	Total Contact Hours:60	Max Marks:100

Course Objectives:

The course is intended to impart knowledge on hands-on experience in building and experimenting with advanced intelligent systems and applications in ai and related technologies.

List of Experiments:

Below listed experiments can be implemented by Spyder

1. Implement Bayesian belief networks
2. Implement hidden markov models
3. Implement EM algorithm for HMM
4. Implement decision problems for various real-world applications
5. Design a Naïve Bayes classifier to classify the given dataset
6. Implement SVM on a Dataset (e.g., Iris dataset) and visualize the decision boundaries
7. Implement gaussian mixture models
8. Implement the reinforcement learning for various reward based applications
9. Implement the value iteration of MDP using the grid world application
10. Mini project

Course Outcomes	Cognitive Level
At the end of this course, students will be able to:	
CO1: Apply probabilistic models such as bayesian networks and hidden markov models to solve uncertainty-based AI problems	Apply
CO2: Analyze the effectiveness of machine learning algorithms like Naïve Bayes, SVM, and GMM on real-world datasets	Analyze
CO3: Evaluate decision-making strategies and reinforcement learning techniques in dynamic environments	Evaluate
CO4: Design an AI-based mini projects using advanced algorithms and tools to address real-time applications	Create

Course Articulation Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	-	-	-	-	-	-	-	-	2	-
CO2	-	-	3	3	-	-	-	-	-	-	-	-	2	-
CO3	-	3	-	-	-	-	-	-	-	-	-	-	-	-
CO4	-	-	-	-	3	2	-	-	2	2	2	1	3	-

High-3; Medium-2; Low-1

Reference Book(s):

R1. Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach",
3rd Edition, Pearson Education, 2021.

Web References:

1.NPTEL Course Content: <http://nptel.ac.in/courses/106106091/>